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from collections import deque

capacity1 = 11
capacity2 = 9
target = 8

visited = set()
queue = deque()

queue.append((0,0,[]))

while queue:

    a,b,path = queue.popleft()

    if a == target or b == target:
        print(path+[(a,b)])
        break

    if (a,b) in visited:
        continue

    visited.add((a,b))

    states = [

        (capacity1,b),
        (a,capacity2),
        (0,b),
        (a,0),

        (a-min(a,capacity2-b),
         b+min(a,capacity2-b)),

        (a+min(b,capacity1-a),
         b-min(b,capacity1-a))
    ]

    for s in states:
        if s not in visited:
            queue.append((s[0],s[1],path+[(a,b)]))

```